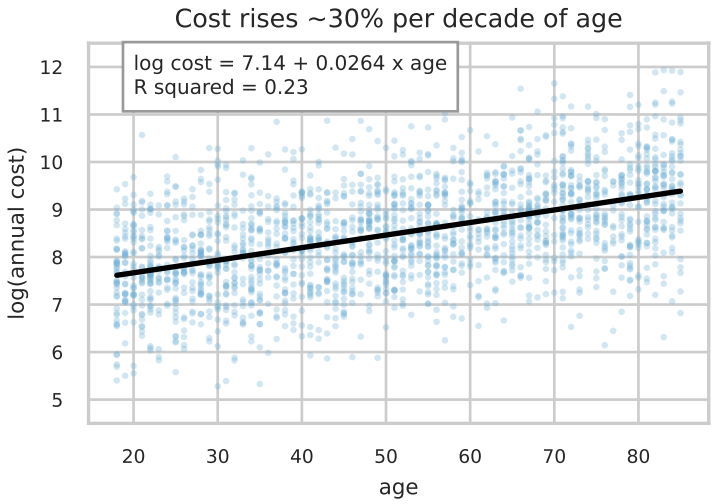
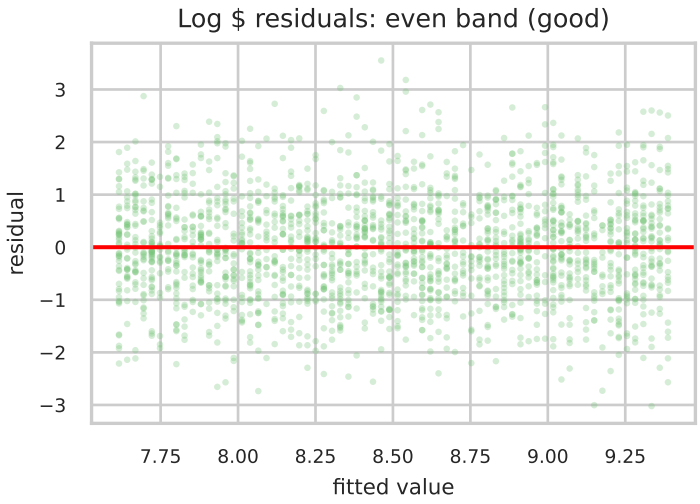
Day 6 Cheat Sheet: Correlation & Simple Linear Regression

Measure the link, fit a line, check the leftovers, then question the cause

TERM	WHAT IT TELLS YOU	CODE
Pearson r	straight-line strength, -1 to +1	stats.pearsonr(x, y)
Spearman rho	same idea on ranks (skew/outlier safe)	stats.spearmanr(x, y)
Slope (b)	change in y for +1 unit of x	stats.linregress(x, y).slope
Intercept	predicted y when x = 0 (often not real)	fit.intercept
R squared	share of y's variation the line explains	fit.rvalue**2 / model.rsquared
Residual	actual - predicted (what the line misses)	model.resid
95% CI, p	how sure you are about the slope	smf.ols("y ~ x", df).fit().summary()

READING A LOG-COST SLOPE: if y = log(cost), slope b means cost changes by (exp(b) - 1) per unit of x.
b = 0.0264 per year -> exp(0.0264) - 1 = 2.7% per year, exp(0.264) - 1 = 30% per decade





THIS DATA (simulated, reused from Day 5): 2,000 patients after dropping 12 duplicates
Age vs. raw cost: Pearson 0.33, Spearman 0.48 | age vs. log cost: Pearson 0.48, Spearman 0.48
log cost = 7.14 + 0.0264 x age (95% CI for slope 0.0243 to 0.0286), R squared = 0.233
Predicted typical cost: age 40 $3,632, age 70 $8,031 | R squared on raw dollars only 0.11
ER visits vs. log cost: r = 0.22, but the coefficient goes from +0.267 to -0.011 (p = 0.63) after adding chronic + smoker

HABITS THAT KEEP YOU HONEST
- Always plot first: the same r can hide a curve, clusters, or one wild outlier
- Skewed cost? Model log(cost), then translate the slope back to a % change
- R squared of 0.23 is not "bad": age is one driver among many. Say what is missing
- Correlation is not causation: ask what common cause (smoking, illness) drives both

```
from scipy import stats; import statsmodels.formula.api as smf
df["log_cost"] = np.log(df["annual_cost"])
stats.pearsonr(df["age"], df["log_cost"]); stats.spearmanr(df["age"], df["log_cost"])
m = smf.ols("log_cost ~ age", data=df).fit(); print(m.summary())
plt.scatter(m.fittedvalues, m.resid); plt.axhline(0)          # residual plot
smf.ols("log_cost ~ er_visits + chronic_conditions + smoker", df).fit().params
```

Infographic: From Scatter Plot to Honest Regression

Simulated: 2,000 patients, regress log(annual cost) on age, then test a tempting causal story

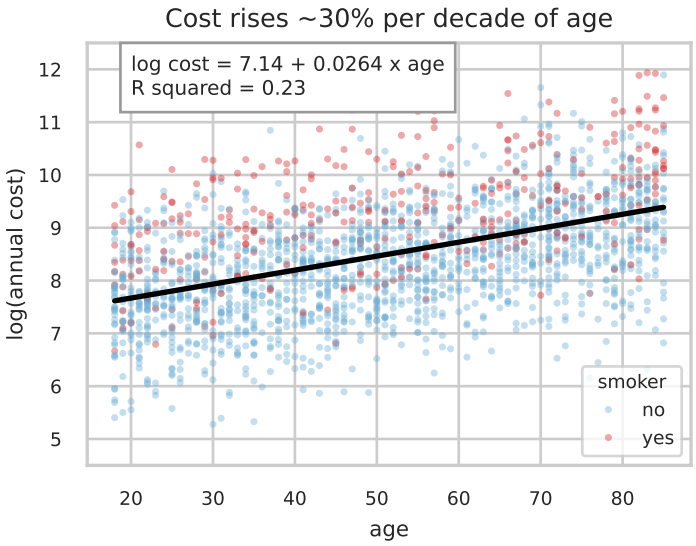
Plot	Correlate	Fit	Translate	Residuals	Question
scatter x vs. y	Pearson and Spearman	OLS line: slope, R sq.	log slope to % change	shapeless cloud?	what else drives both?

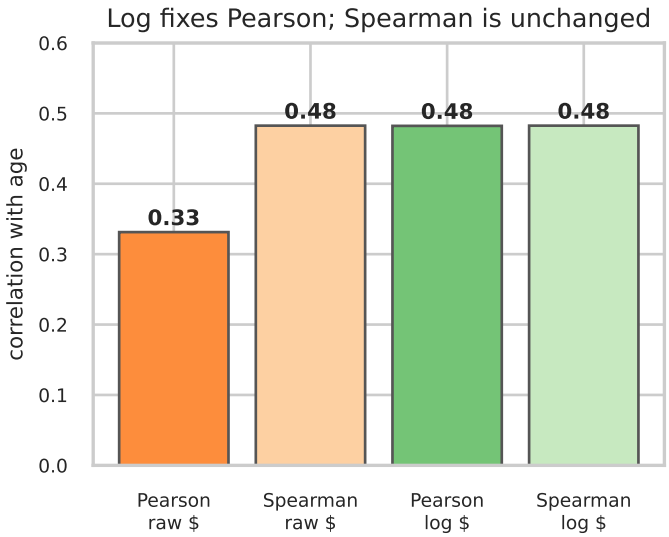
PEARSON r
0.33 raw
0.48 log

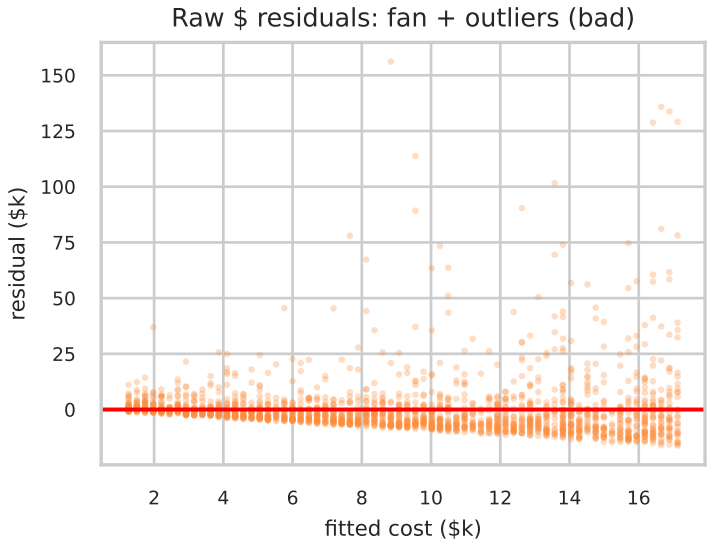
R SQUARED
0.23
age alone

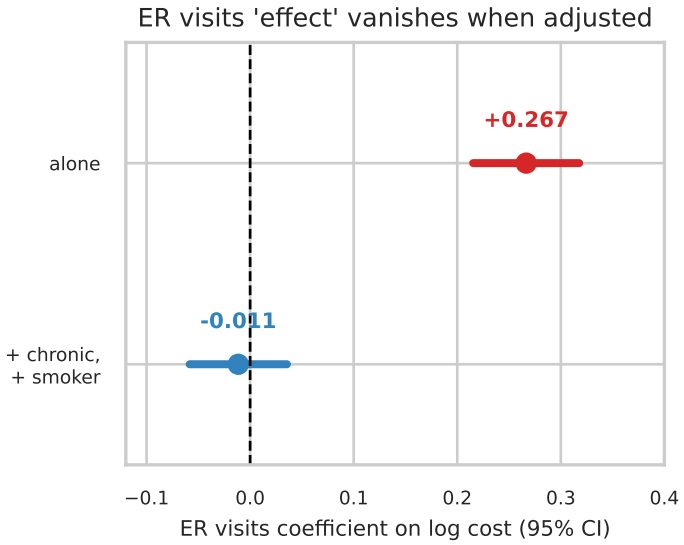
PER DECADE
+30%
cost

ER VISITS
+0.27 to
-0.01 adj.









Causation trap: patients with more ER visits cost more (r = 0.22). But smoking and chronic illness drive BOTH. Once they are in the model, the ER coefficient is -0.011 (95% CI -0.058 to 0.036). In this simulation, ER visits never caused cost.

Hands-On Worksheet: ds-genai-90-days / day06

60-90 minutes | notebook: day06/regression_patients.ipynb

[ ] mkdir day06, copy day05/patients.csv in, conda activate ds90

[ ] conda install statsmodels scikit-learn (if not already in ds90)

[ ] Load, drop duplicates (2,000 rows), add log_cost = np.log(annual_cost)

[ ] Pearson and Spearman: age vs. raw cost and vs. log cost. Why does only Pearson move?

[ ] stats.linregress(age, log_cost): write down slope, intercept, R squared

[ ] Translate the slope: exp(b) - 1 per year and exp(10b) - 1 per decade

[ ] smf.ols('log_cost ~ age').fit().summary(): find coef, std err, 95% CI, R sq.

[ ] Check sklearn LinearRegression gives the same slope; predict ages 40 and 70

[ ] Residual plots: raw dollars vs. log cost. Which one is a shapeless cloud?

[ ] Causation test: er_visits alone, then + chronic_conditions + smoker

[ ] Add a findings cell; git add day06 && git commit && git push

[ ] Stretch: Snowflake CORR, REGR_SLOPE, REGR_R2 on LN(annual_cost)

```
import numpy as np, pandas as pd, statsmodels.formula.api as smf
from scipy import stats
df = pd.read_csv("patients.csv").drop_duplicates()
df["log_cost"] = np.log(df["annual_cost"])
print(stats.pearsonr(df["age"], df["log_cost"]), stats.spearmanr(df["age"], df["log_cost"]))
m = smf.ols("log_cost ~ age", data=df).fit()
print(m.params, m.rsquared, m.conf_int())        # expect slope ~0.0264, R sq. ~0.233
print(np.expm1(10 * m.params["age"]))             # % change per decade
adj = smf.ols("log_cost ~ er_visits + chronic_conditions + smoker", data=df).fit()
print(adj.params["er_visits"], adj.pvalues["er_visits"])
```

Pearson raw: _____ log: _____ Spearman raw: _____ log: _____ Slope: _____

R squared: _____ % per decade: _____ ER coef alone: _____ adjusted: _____

Day 6 Self-Check Quiz

1. Pearson r between age and raw cost is 0.33, but 0.48 with log cost. Why the jump?

_____

2. Why does Spearman rho stay at 0.48 whether you use raw or log cost?

_____

3. Slope of log cost on age is 0.0264. Say it in plain words for a manager.

_____

4. R squared is 0.23. What does that mean, and is the model useless?

_____

5. What should a healthy residual plot look like? Name one warning sign.

_____

6. The intercept is 7.14 (cost about $1,260 at age 0). Should you report it?

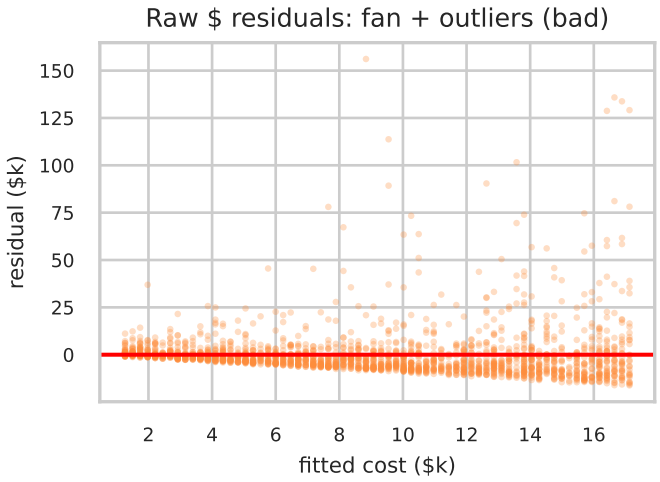
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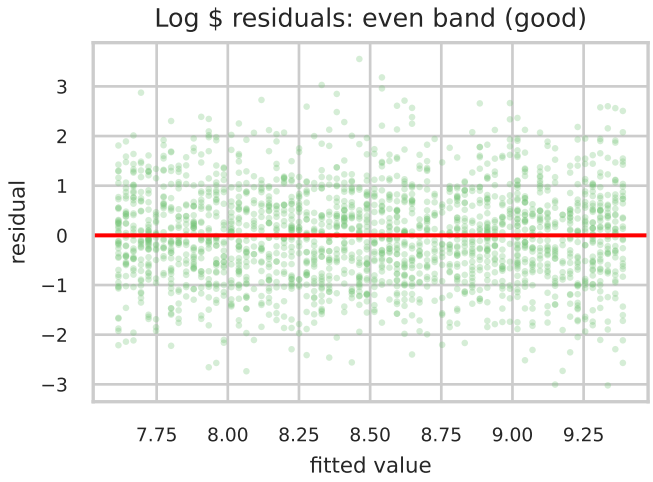
7. ER visits predict higher cost (r = 0.22). Does cutting ER visits cut cost here?

_____

8. When would you pick sklearn LinearRegression over statsmodels OLS?

_____





Answers:
1 Pearson measures straight-line fit; the long right tail of raw cost bends and stretches it. Log makes it closer to a line
2 Spearman uses ranks, and log never changes the order of values 3 "Each extra year of age goes with about 2.7% higher cost, about 30% per decade"
4 age explains about 23% of the variation in log cost; the rest is smoking, illness, chance. Useful, just incomplete
5 an even, shapeless band around 0; warning signs: fan shape, curve, clusters, extreme outliers
6 not as a finding: nobody in the data is under 18, so age 0 is outside the range 7 no: after adding chronic + smoker its coefficient is -0.011, p = 0.63
8 when the goal is prediction inside a scikit-learn pipeline; statsmodels when you need p-values, CIs and a summary